

Population-level prediction of depression, Alzheimer's disease, and Parkinson's disease incidence using artificial intelligence: implications for health service planning

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01 / BACKGROUND

We know the past. Can we anticipate future cases? Weekly bulletins from Mexico's National Epidemiological Surveillance System [1] compile new cases reported by health services. We forecast **notifications, not clinical incidence**.

02 / OBJECTIVES

Forecast **52 weeks** of notified **depression (F32), Alzheimer's (G30) and Parkinson's (G20)** cases, conditions of neuropsychiatric interest. Measure error to explore potential use in planning services, staffing and budgets.
ICD-10: International Classification of Diseases, tenth revision.

03 / METHODS

Retrospective study, 2014-2026. Time series track weekly cases by condition, area and sex.
333 series = 3 conditions × 37 areas × 3 strata
Areas: 32 states, the country and 4 state groups.
Strata: men, women and both sexes combined.
Results: both sexes combined; three conditions nationally and depression in 32 states. National forecasts are not the sum of state forecasts.

How do we forecast?

Four statistical and **machine learning (ML)** methods: **Prophet** [2] models trends and cycles; **DeepAR** [3] learns through a neural network. **Ensemble** combines forecasts; **Stacking** learns to combine them. A model is selected per series: nationally, DeepAR for depression and Prophet for Parkinson's and Alzheimer's.

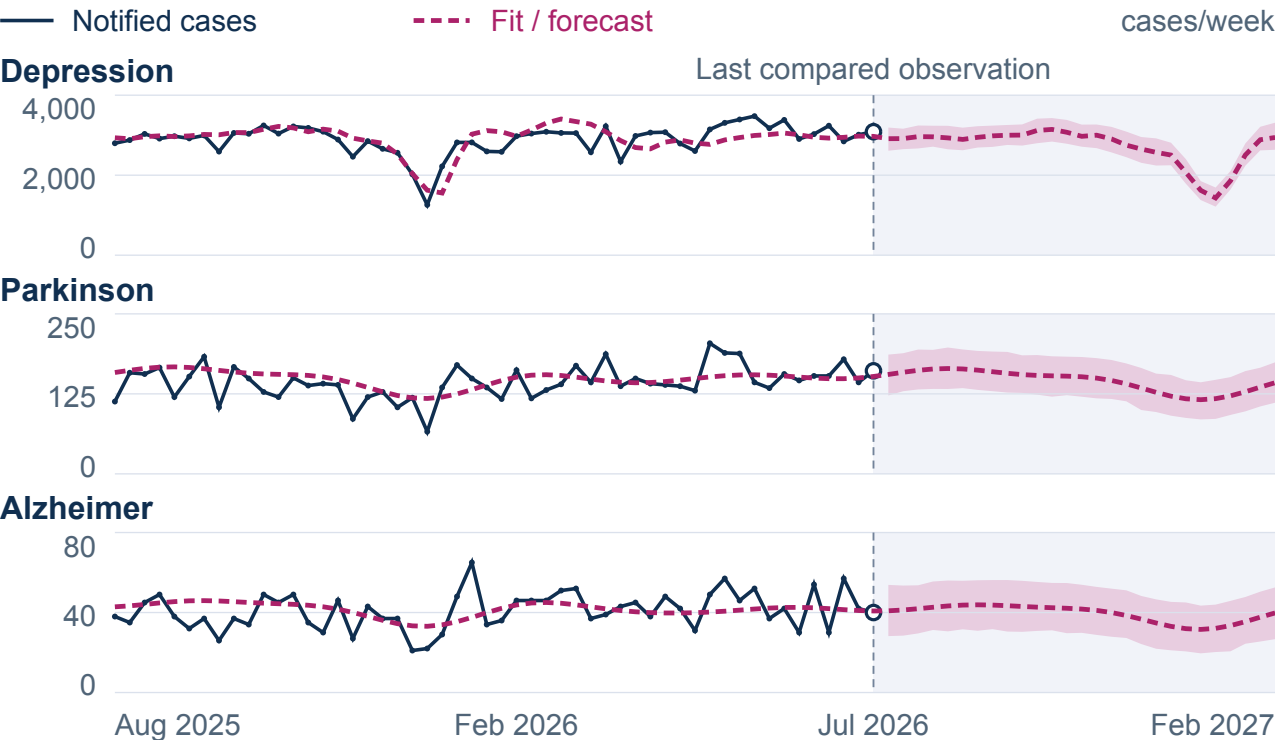
04 / EVALUATION & RESULTS

We evaluated forecasts against 2026 notifications: **epidemiological weeks 7-31 (25 weeks)**. For **24-31 (eight weeks)**, predictions were saved in advance. Two metrics assess forecast performance: **sMAPE** (symmetric mean absolute percentage error): compares relative error across series; lower is better, 0% is an exact match. **MAE** (mean absolute error): average error in cases/week, without cancelling signs.

Table 1. National errors

Condition	sMAPE		MAE	
	25 weeks	8 weeks	25 weeks	8 weeks
Depression	8.7%	5.0%	260.2	153.8
Parkinson	10.6%	6.7%	16.5	10.3
Alzheimer	14.4%	17.8%	6.2	7.4

Figure 1. History and forecast



History: retrospective fit. Model band; coverage not evaluated.

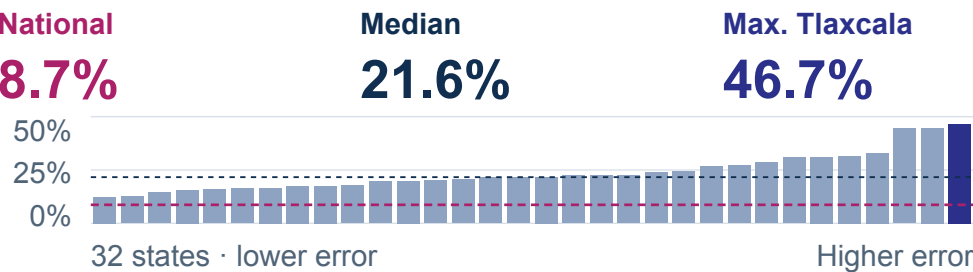
DEPRESSION · NATIONAL TOTAL

-1.6% below the reported total

74,675 Forecast | **75,859** Notified

Difference in 25-week totals, not weekly error.

Figure 2. Depression · sMAPE, 25 weeks



National accuracy does not guarantee local accuracy. With few cases, small errors can be relatively large.

ANOTHER PERSPECTIVE · DEPRESSION BY SEX

Cumulative 2026 notifications through week 31.
Women 67,753 · 73.1% | Men 24,974 · 26.9%
Different window and variable: totals not comparable with the national callout. Description, not forecasts or individual risk.

05 / CONCLUSIONS

Preparing decisions. Forecasts could inform staffing and appointments, not determine budgets. They require local prospective validation; **usefulness for services has not yet been evaluated.**

LIMITATIONS. No measured care demand or clinical benefit; not fully prospective. No age or external-country evaluation; possible underreporting.

[1] DGE. Boletín Epidemiológico, SINAVE. Secretaría de Salud, México; 2026.
[2] Taylor SJ, Letham B. Forecasting at scale. Am Stat. 2018;72:37-45.
[3] Salinas D, et al. DeepAR. Int J Forecast. 2020;36:1181-1191.



Guided by compassion, grounded in science: psychiatry for our time.



Methods & data
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